

Mathematical Principles of Cognitive Systems

Author: Heng Zhao (赵恒)

Affiliation: Independent Researcher

Email: zhaoheng1990@gmail.com

Subtitle: Possibility Spaces, Constraint Fields, and Cbit Compression

Date: 2026-06-08

Abstract

A cognitive system is not merely an answer generator, a symbol manipulator, or a message channel. This paper proposes a mathematical framework for general cognitive systems by defining the cognitive ontology as a coupled possibility-space and constraint-field structure:

$$\mathcal{P}_c = (\Omega_c, \mathcal{F}_c)$$

where Ω_c is a possibility space composed of candidate cognitive structures, and $\mathcal{F}_c$ is a constraint field acting on that space. A candidate cognitive structure is defined as:

$$S_i = (O_i, R_i, C_i, \Gamma_i)$$

where O_i denotes a set of objects, R_i a set of relations, C_i a set of constraints, and Γ_i a family of expandable trajectories. We further define cognitive entropy as:

$$H_c = - \sum_i p_i \log_2 p_i$$

and define a Cbit as the unit of effective compression of the possibility space of candidate cognitive structures:

$$\Delta Cbit = H_c^{before} - H_c^{after}$$

The central claim is that cognition is not answer generation, symbolic processing, or message transmission. Cognition is effective Cbit compression of a possibility space under a constraint field. Under this view, learning, understanding, reasoning, memory, creativity, misunderstanding, hallucination, scientific research, and education can be interpreted as different forms of possibility-space compression. Information theory is treated as a low-dimensional projection of this framework onto message space. Large language models are included only as an interpretive example and a prediction interface; no unpublished empirical evidence about LLMs is reported in this paper.

1. Introduction: Why Cognitive Systems Need a New Mathematical Formulation

A cognitive system is not merely a machine that produces answers.

When a cognitive agent faces a question, an object, an environment, or a task, it does not first face a predefined set of answers. It faces an incompletely organized possibility space. An answer is only the terminal manifestation of a cognitive process. The real process occurs before the answer appears: the system must rank possible objects, possible relations, possible constraints, and possible explanatory trajectories; eliminate unstable candidates; compress

ineffective expansions; and gradually commit to a dominant trajectory.

Existing theories often understand cognition from three perspectives.

First, cognition is treated as answer generation. A question enters the system, and an answer comes out. This view is convenient for engineering, but it reduces cognition to an input-output map and ignores the candidate space, constraint selection, trajectory competition, and structural compression that precede the answer.

Second, cognition is treated as symbolic processing. The system receives symbols, manipulates symbols, combines symbols, and derives conclusions through rules. This view explains parts of logical reasoning, but it has difficulty explaining intuition, transfer, creativity, misunderstanding, hallucination, and frame revision in ambiguous settings.

Third, cognition is treated as information transmission. The system receives, encodes, transmits, and decodes messages. Information theory provides a powerful mathematical theory of message uncertainty, but cognitive systems usually do not face messages alone. They face candidate cognitive structures composed of objects, relations, constraints, and expandable trajectories. Message uncertainty is only a surface-level projection of cognitive uncertainty.

This paper proposes a more general mathematical formulation:

$$\boxed{\textit{Cognition} = \textit{effective Cbit compression of a possibility space under a constraint field.}}$$

The possibility space is not a metaphor. It is the set of candidate cognitive structures that the system can unfold, compare, and compress. The constraint field is not a simple rule set. It is a multidimensional structure of viability, closure, consistency, compressibility, transferability, and cost acting on those candidates.

Thus, the central object of this paper is not an answer, not a symbol, and not a message. It is:

$$\boxed{\mathcal{P}_c = (\Omega_c, \mathcal{F}_c)}$$

where Ω_c is a possibility space and $\mathcal{F}_c$ is a constraint field.

The contributions of this paper are fourfold.

First, it introduces a general cognitive ontology $\mathcal{P}_c = (\Omega_c, \mathcal{F}_c)$.

Second, it defines a candidate cognitive structure $S_i = (O_i, R_i, C_i, \Gamma_i)$, so that cognitive objects are not reduced to answers or symbols.

Third, it defines cognitive entropy and Cbit, treating cognition as effective compression of possibility space.

Fourth, it shows how information theory, learning, understanding, reasoning, memory, creativity, misunderstanding, and scientific research can be interpreted as projections or consequences of this framework.

This is not a paper about LLM mechanisms and does not report unpublished LLM empirical evidence. LLMs appear only as an interpretive example and as a future prediction interface.

2. Cognitive Ontology: Possibility Space and Constraint Field

2.1 Cognition Faces a Possibility Space

When a cognitive system faces a problem q , it does not directly face an answer a . It faces a possibility space composed of candidate cognitive structures:

$$\Omega_c = \{S_i\}_{i \in I}$$

where each S_i is a candidate cognitive structure.

Three levels must be distinguished.

The first level is the possibility space Ω_c . It is the global ontological domain faced by the cognitive system.

The second level is the candidate cognitive structure S_i . It is an element of the possibility space.

The third level is structural organization: the internal organization of objects, relations, constraints, and trajectories within S_i .

Therefore, this paper does not call Ω_c a “structure space.” The ontology is the possibility space. Structure is the internal organization of candidate elements within that space. This distinction is crucial. If the ontology is called a structure space, the theory drifts toward static structuralism. The present theory concerns the dynamic compression of open possibilities under a constraint field.

2.2 Candidate Cognitive Structures

A candidate cognitive structure is defined as:

$$S_i = (O_i, R_i, C_i, \Gamma_i)$$

where O_i denotes a set of objects, R_i a set of relations, C_i a set of constraints, and Γ_i a family of expandable trajectories.

This definition means that a cognitive candidate is not a simple answer. It is an expandable structural unit. It contains what the objects are, how they are related, what constrains their relations, and what trajectories can unfold from that structure.

Incorrect candidate structures may also exist in Ω_c . They may contain incorrect relations, incorrect object bindings, or incorrect constraints, while still maintaining local closure. This explains why cognitive errors are often not structureless. They are structured candidates that have gained abnormal advantage under the constraint field.

2.3 Constraint Field

A possibility space alone is not sufficient for cognition. An unconstrained possibility space is merely an unorganized set of candidates. A cognitive system must judge which candidates are more viable, more consistent, more stable, less costly, and more transferable.

We define the constraint field as:

$$\mathcal{F}_c : \Omega_c \rightarrow \mathbb{R}^k$$

For each candidate cognitive structure S_i , the constraint field assigns a multidimensional evaluation:

$$\mathcal{F}_c(S_i) = (P_i, LC_i, CL_i, CP_i, TR_i, Cost_i)$$

where P_i denotes plausibility, LC_i logical consistency, CL_i closure, CP_i compression efficiency, TR_i transferability, and $Cost_i$ cognitive cost.

In some settings, the constraint field can be compressed into a scalar potential:

$$U_c(S_i) = -\alpha P_i - \beta LC_i - \gamma CL_i - \delta CP_i - \eta TR_i + \rho Cost_i$$

A lower $U_c(S_i)$ means that the candidate is more likely to become a dominant trajectory.

This does not imply that cognitive systems explicitly compute such a potential. It is a mathematical abstraction of how the constraint field changes the relative viability of candidates.

2.4 Coupling Between Possibility Space and Constraint Field

Ω_c and $\mathcal{F}_c$ cannot be understood separately.

With Ω_c but without $\mathcal{F}_c$, the system can list possibilities but cannot determine which ones should be retained, rejected, or further unfolded.

With $\mathcal{F}_c$ but without Ω_c , the constraint field has no domain.

Therefore, the cognitive ontology must be:

$$\mathcal{P}_c = (\Omega_c, \mathcal{F}_c)$$

This is a coupled dual structure. Ω_c gives what can be unfolded. $\mathcal{F}_c$ gives what can remain viable. Cognition occurs through their coupling.

3. Cognitive Entropy: Uncertainty Over Candidate Structures

3.1 Probabilities Over Candidate Structures

Given a problem, task, or input q , the cognitive system assigns weights to candidate cognitive structures:

$$p_i = P(S_i \mid q, \mathcal{F}_c)$$

Here p_i need not be an explicitly represented subjective probability. It may represent attention weight, trajectory advantage, behavioral tendency, internal state weight, human availability, or any general measure of effective competition among candidates.

The essential condition is:

$$\sum_i p_i = 1$$

and p_i expresses the relative effectiveness of candidate S_i in the current cognitive state.

3.2 Cognitive Entropy

Cognitive entropy is defined as:

$$H_c(q) = - \sum_i p_i \log_2 p_i$$

Cognitive entropy measures uncertainty over candidate cognitive structures, not uncertainty over message symbols.

When many candidates are effectively competing, cognitive entropy is high. When the system eliminates ineffective candidates, strengthens some candidates, and eventually commits to a dominant trajectory, cognitive entropy decreases.

3.3 Effective Number of Possibilities

The effective number of possibilities is defined as:

$$N_{eff} = 2^{H_c}$$

If $H_c = 3$, then $N_{eff} = 8$. This does not necessarily mean that the system literally has eight discrete candidates. It means that the effective complexity of the current distribution is equivalent to eight uniformly competing candidates.

3.4 Cognitive Gain

Cognitive gain is defined as:

$$G_c = H_c^0 - H_c^T$$

If $G_c > 0$, the system has compressed its possibility space.

However, $G_c > 0$ alone does not imply cognitive success. Incorrect compression, biased compression, and premature collapse may also reduce entropy. A quality term is therefore required.

4. Cbit: Cognitive Bit

4.1 Definition

A Cbit is a cognitive bit:

$$Cbit = \text{Cognitive bit}$$

It is the unit of effective compression of the possibility space of candidate cognitive structures:

$$\Delta Cbit = H_c^{before} - H_c^{after} = \log_2 \frac{N_{eff}^{before}}{N_{eff}^{after}}$$

When $\Delta Cbit = 1$, the effective possibility space has been reduced by half.

4.2 Cbit Versus Bit

A bit in information theory measures the elimination of message uncertainty. A Cbit measures the effective compression of a space of candidate cognitive structures.

The objects differ.

A bit concerns a message variable:

$$X \in \{x_1, x_2, \dots, x_n\}$$

A Cbit concerns a candidate cognitive structure:

$$S_i = (O_i, R_i, C_i, \Gamma_i)$$

Thus, a bit is a unit of message selection. A Cbit is a unit of cognitive-structure selection.

A bit answers: which symbol was transmitted?

A Cbit answers: how much of the candidate cognitive structure space was excluded, and what effective structure remained?

4.3 Effective Cbit

Not every entropy reduction is effective cognition.

If the system randomly deletes half of its candidates, entropy decreases, but cognition has not improved.

If the system prematurely eliminates the correct candidate, entropy also decreases, but the compression is wrong.

If the system retains a locally coherent but globally false candidate structure, entropy may decrease while misunderstanding or hallucination is produced.

We therefore define effective Cbit:

$$Cbit_{eff} = \Delta H_c \cdot Q_c$$

where Q_c is a cognitive quality term:

$$Q_c = f(Closure, Consistency, Compression, Transferability, Cost, TaskFitness)$$

If the compressed structure is more closed, consistent, transferable, efficient, and task-adapted, then $Q_c > 0$. If the compression is produced by incorrect constraints, bias, premature commitment, or wrong closure, then $Q_c \leq 0$.

High-quality cognition is therefore not merely a reduction in H_c , but an increase in $Cbit_{eff}$.

4.4 Cbit-Gain Criterion

A general criterion follows:

$$\Delta Cbit_{eff} > 0$$

This applies not only to individual cognitive processes, but also to theory construction, experiment design, memory retention, and frame revision.

Is a new concept worth introducing? Is a new variable necessary? Is an experiment useful? Should a memory be retained? Should a framework continue to be patched? These questions can be reframed as: does it produce future effective Cbit gain?

5. Cognitive Dynamics

5.1 Cognition as Trajectory

If a cognitive system faces a possibility space, then a cognitive process can be treated as a trajectory in that space.

Define a cognitive trajectory:

$$\gamma : [0, T] \rightarrow \Omega_c$$

where $\gamma(t)$ denotes the distributional state of candidate structures at time or inference step t .

$\gamma(t)$ need not be a single definite candidate. It may be a probability distribution, a weight distribution, or a state of effective competition over candidate structures.

Thus, cognition is not:

$$q \rightarrow a$$

but rather:

$$q \rightarrow \Omega_c^0 \rightarrow \Omega_c^1 \rightarrow \dots \rightarrow \Omega_c^T \rightarrow a$$

The answer a is a terminal projection of the trajectory, not the cognitive ontology.

5.2 Cognitive Action

To evaluate trajectories, define cognitive action:

$$\mathcal{A}_c(\gamma) = \int_0^T L(S_t, \dot{S}_t, \mathcal{F}_c) dt$$

where $L(S_t, \dot{S}_t, \mathcal{F}_c)$ is the local cognitive cost under state S_t , transition rate $\dot{S}_t$, and constraint field $\mathcal{F}_c$.

This cost may include structural transition cost, constraint conflict, memory access cost, wrong-closure risk, resource consumption, trajectory instability, and transfer failure.

A dominant cognitive trajectory can be written as:

$$\gamma_* \approx \arg \min_{\gamma} \mathcal{A}_c(\gamma)$$

This does not imply that cognitive systems explicitly solve a variational problem. It means that high-quality cognitive trajectories tend to be low-cost and constraint-consistent.

5.3 Effective Cbit Rate

Because cognitive systems have limited resources, absolute Cbit gain is less important than effective Cbit gain per unit cost.

Define:

$$\mathcal{R}_c(\gamma) = \frac{\Delta Cbit_{eff}(\gamma)}{Cost(\gamma)}$$

The dominant trajectory can then be expressed as:

$$\gamma_* \approx \arg \max_{\gamma} \frac{\Delta Cbit_{eff}(\gamma)}{Cost(\gamma)}$$

High-quality cognition is not necessarily the shortest path or the path with the largest amount of information. It is the path with the highest effective Cbit gain per unit resource.

5.4 Near-Geodesic Cognition

If the possibility space has a geometric structure, cognitive trajectories can be interpreted as near-geodesic paths.

Here, however, “geodesic” should not mean simply the shortest path in an abstract geometry.

In cognitive systems, a near-geodesic is:

a low – action path that maximizes effective Cbit gain per unit resource in a constrained possibility space.

Correct cognition often appears near-geodesic not because the system mechanically seeks shortest paths, but because high-quality compression avoids useless detours, wrong closures, ineffective branches, and costly reversals.

6. Learning, Understanding, Reasoning, Memory, and Creativity

6.1 Learning

Learning is long-term reusable effective Cbit gain:

$$\boxed{Learning = \Delta Cbit_{eff}^{long-term}}$$

A temporary judgment may reduce current cognitive entropy, but it is not necessarily learning. It becomes learning only if the compression capability can be reused in future tasks.

Learning is therefore not the storage of more information. It improves future compression by lowering initial cognitive entropy, increasing Q_c , or increasing $\Delta Cbit_{eff}/Cost$.

6.2 Understanding

Understanding is not memorizing a definition or repeating an answer.

Understanding means that a concept is no longer isolated. It has formed high-quality connections in the cognitive possibility space.

When a concept is isolated, it activates many unstable candidates, producing high H_c . When a concept is understood, related objects, relations, constraints, and trajectories become organized, allowing the system to eliminate ineffective expansions and form stable trajectories. Thus N_{eff} decreases.

Understanding can be defined as:

the formation of reusable constraint structures that reduce future cognitive entropy.

6.3 Reasoning

Reasoning is the competition, selection, transformation, and commitment among candidate cognitive structures.

In this framework, reasoning is not a linear chain from premise to conclusion. It is:

$$\boxed{Reasoning : \Omega_c^0 \rightarrow \Omega_c^1 \rightarrow \dots \rightarrow \Omega_c^T}$$

It includes candidate generation, constraint filtering, trajectory competition, local closure checks, elimination of wrong paths, and commitment to a dominant trajectory.

6.4 Memory

Memory is not the storage of the past.

Memory changes the initial state or unfolding rule of future possibility spaces.

If a memory is effective, it lowers future H_c^{init} , raises Q_c , or lowers $Cost$.

Thus, the function of memory is not to preserve content but to preserve structures that can regenerate effective cognitive trajectories.

6.5 Creativity

Creativity is often confused with divergence.

Pure divergence is not creativity. It merely enlarges the possibility space.

True creativity contains two stages.

First, it expands or reconstructs the possibility space:

$$\Omega_c \rightarrow \Omega'_c$$

Second, it forms a new constraint-field organization:

$$\mathcal{F}_c \rightarrow \mathcal{F}'_c$$

A new idea that only increases possibilities without enabling higher-quality compression is noise. A new framework that enlarges exploration while producing higher $Cbit_{eff}$ is creativity.

Therefore:

Creativity = possibility expansion followed by higher – order effective compression.

7. Error, Misunderstanding, and Hallucination

7.1 Error Is Not Absence of Compression

In this framework, erroneous cognition is not the absence of compression.

Many wrong judgments, misunderstandings, and hallucinations have strong local structure. They are often not random. They form local closure while violating global constraints.

Thus:

$$\Delta H_c > 0$$

but:

$$Q_c < 0$$

so:

$$Cbit_{eff} < 0$$

The system has compressed the possibility space, but in the wrong direction.

7.2 Misunderstanding

Misunderstanding is low-quality or negative-quality compression:

Misunderstanding = compression under an incorrect or incomplete constraint field.

7.3 Hallucination

Hallucination can be generally defined as:

Hallucination = wrong but locally closed cognitive trajectory wins.

This definition applies not only to artificial systems but also to human cognition.

Many hallucinated judgments are not completely chaotic. They often contain local logic, semantic closure, and local explanatory force, while violating higher-level constraints.

Hallucination is therefore not random error. It is the abnormal dominance of a wrong closure trajectory under the constraint field.

7.4 Cognitive Inertia

Cognitive inertia occurs when a system continues to unfold within an old possibility space or an old constraint field, even though its effective Cbit has declined.

Overcoming cognitive inertia requires either switching the object or switching the unfolding mode.

8. Information Theory as a Projection of Cognitive Theory

8.1 Message Projection

Define a projection from cognitive possibility space to message space:

$$\pi_m : \Omega_c \rightarrow \Omega_m$$

where:

$$x_j = \pi_m(S_i)$$

is the projection of candidate cognitive structure S_i into message space.

Multiple candidate structures may project to the same message:

$$p(x_j) = \sum_{S_i: \pi_m(S_i)=x_j} p(S_i)$$

8.2 Information Entropy

Information entropy is:

$$H_m = - \sum_j p(x_j) \log_2 p(x_j)$$

Because π_m is lossy:

$H_m \leq H_c$

Message uncertainty is therefore less than or equal to cognitive-structure uncertainty.

8.3 Position of Information Theory

Information theory can be repositioned as:

Information Theory = Message – space projection of Cognitive Theory.

This does not reject information theory. Information theory is foundational for communication systems.

But cognitive systems do more than transmit messages. They construct, compress, select, and reconstruct candidate cognitive structures.

8.4 Boundary Between Communication and Cognition

The central problem of communication is reliable message transmission through a noisy channel.

The central problem of cognition is effective compression of a possibility space under a constraint field.

Communication optimizes message transfer. Cognition optimizes structural compression.

Thus:

Message uncertainty reduction is a projection of cognitive possibility – space compression.

9. Mathematical Formalization Audit

9.1 Domain Audit

The first mathematical requirement is a clear domain:

$$\Omega_c = \{S_i\}_{i \in I}$$

where:

$$S_i = (O_i, R_i, C_i, \Gamma_i)$$

This definition avoids reducing cognition to answers, allows candidates to have internal organization, and allows candidates to be generated, split, merged, and eliminated.

9.2 Constraint-Field Audit

The constraint field is:

$$\mathcal{F}_c : \Omega_c \rightarrow \mathbb{R}^k$$

Its domain is Ω_c , and its range is a multidimensional evaluation.

This ensures that the constraint field is not a metaphor: it has a clear object of action.

9.3 Probabilization Audit

Cognitive entropy requires a weight distribution over candidate structures:

$$p_i = P(S_i \mid q, \mathcal{F}_c)$$

The theory does not require the system to explicitly represent probabilities. It only requires candidates to have comparable effective weights.

9.4 Cbit Audit

Cbit is defined as:

$$\Delta Cbit = H_c^{before} - H_c^{after}$$

This inherits additivity from entropy differences while raising the object of compression from message variables to candidate cognitive structures.

Effective Cbit is:

$$Cbit_{eff} = \Delta H_c \cdot Q_c$$

which prevents wrong compression from being counted as cognitive success.

9.5 Dynamical Audit

The cognitive trajectory is:

$$\gamma : [0, T] \rightarrow \Omega_c$$

and the cognitive action is:

$$\mathcal{A}_c(\gamma) = \int_0^T L(S_t, \dot{S}_t, \mathcal{F}_c) dt$$

The framework therefore describes not only static candidate sets, but also path selection, transition cost, and dynamic constraint interaction.

9.6 Information-Theoretic Projection Audit

The message projection is:

$$\pi_m : \Omega_c \rightarrow \Omega_m$$

which yields:

$$H_m \leq H_c$$

Thus, information theory is not rejected. It is placed at the projection level.

9.7 Minimal Axiom System

Axiom 1: Possibility Ontology

$$\Omega_c = \{S_i\}$$

Cognition faces a possibility space rather than an answer set.

Axiom 2: Candidate Cognitive Structure

$$S_i = (O_i, R_i, C_i, \Gamma_i)$$

Candidate possibilities are organized as objects, relations, constraints, and expandable trajectories.

Axiom 3: Constraint Field

$$\mathcal{F}_c : \Omega_c \rightarrow \mathbb{R}^k$$

Candidate structures are acted upon by a constraint field.

Axiom 4: Cognitive Entropy

$$H_c = - \sum_i p_i \log_2 p_i$$

Cognitive uncertainty is defined over candidate cognitive structures.

Axiom 5: Cbit Compression

$$\Delta Cbit = H_c^{before} - H_c^{after}$$

Cognitive gain appears as possibility-space compression.

Axiom 6: Effective Cognition

$$Cbit_{eff} = \Delta H_c \cdot Q_c$$

Only high-quality compression counts as effective cognition.

Axiom 7: Cognitive Dynamics

$$\gamma : [0, T] \rightarrow \Omega_c$$

Cognition is trajectory dynamics in possibility space.

Axiom 8: Message Projection

$$\pi_m : \Omega_c \rightarrow \Omega_m$$

Message transmission is a projection of cognition.

10. LLM Interpretation and Prediction Interface

This section provides only an interpretive mapping and a set of testable predictions. It does not report any LLM empirical results, and no LLM variable is used to define the ontology of this paper.

Large language models may be interpreted as one artificial realization domain of the theory. In a weak correspondence, input text triggers local initial conditions; model parameters provide a form of semantic-relational geometry; internal states may be viewed as projections of candidate possibility distributions; and output sequences are message-space projections of dominant cognitive trajectories.

Thus an LLM may be interpreted as:

$$Prompt \rightarrow \Omega_c^0 \rightarrow \mathcal{F}_c \rightarrow \gamma_* \rightarrow Message$$

This is an interpretive mapping, not an evidential basis for the theory.

The framework yields several predictions.

Prediction 1: If LLM reasoning can be interpreted as possibility-space compression, then before answer output there should exist internal proxy quantities that reflect a transition from high to low uncertainty over candidate explanations.

Prediction 2: If Cbit is the compression unit of candidate cognitive structure space, then high-quality outputs should correspond to higher effective internal compression rates, not merely higher final answer probabilities.

Prediction 3: If a prompt is not an answer instruction but a generator of local initial conditions, then different prompts should alter the unfolding of the initial possibility space, not only the final answer distribution.

Prediction 4: If erroneous output is not random error but the dominance of a wrong closure trajectory, then LLM hallucinations should exhibit local structure. Wrong answers should often arise from locally coherent but globally mis-constrained candidate trajectories.

Prediction 5: If the goal of cognitive control is not to directly control answers but to improve effective Cbit compression, then effective intervention in LLMs should modify the relative advantage of candidate trajectories rather than merely increasing the probability of a chosen answer token.

Prediction 6: If memory lowers future possibility space or improves unfolding, then LLM long-term memory should be evaluated by whether it lowers future initial candidate space, improves correct trajectory initialization, reduces wrong closure, or improves future $\Delta Cbit_{eff}/Cost$, rather than by whether it stores text.

These predictions do not depend on this paper reporting any LLM experiment. They are theoretical interfaces that can be tested in a separate empirical paper.

11. Predictions and Research Program

The theory generates testable predictions across human learning, scientific research, education, and artificial cognitive systems.

11.1 Prediction 1: High-Quality Cognition Compresses Possibility Space

If cognition is effective Cbit compression, then high-quality cognition should show:

$$N_{eff}^{init} > N_{eff}^{final}$$

or:

$$H_c^{init} > H_c^{final}$$

More importantly:

$$Cbit_{eff} > 0$$

The possibility space must not only shrink; the remaining structure must have higher closure, consistency, transferability, and task fitness.

11.2 Prediction 2: Error May Reduce Entropy but Produce Negative Effective Cbit

Many errors are not failures to compress. They are wrong compressions:

$$\Delta H_c > 0$$

but:

$$Q_c < 0$$

therefore:

$$Cbit_{eff} < 0$$

11.3 Prediction 3: Understanding Reduces Future Initial Cognitive Entropy

If understanding is the formation of reusable constraint structures, then after understanding a concept, the initial entropy for related tasks should decrease:

$$Understanding(C) \Rightarrow H_c^{init}(q_{related}) \downarrow$$

and:

$$Q_c(q_{related}) \uparrow$$

11.4 Prediction 4: Creativity Has Expansion and Recompression Stages

True creativity should involve two stages:

$$\Omega_c \rightarrow \Omega'_c$$

followed by:

$$H_c(\Omega'_c) \downarrow$$

with:

$$Cbit_{eff} \uparrow$$

Creativity first produces a new candidate space, then compresses it through a new constraint organization.

11.5 Prediction 5: Memory Value Lies in Future Cbit Gain

The value of memory should not be measured by storage volume, but by future gain:

$$Value(M) = \mathbb{E}[\Delta Cbit_{eff}^{future} \mid M] - Cost(M) - Risk(M)$$

A short memory that substantially lowers future entropy is valuable. A large record that does not improve future compression is not.

11.6 Prediction 6: Scientific Theories Compress Hypothesis Spaces

A scientific theory is not a list of facts. It is a hypothesis-space compressor.

A strong theory should reduce:

$$H_c(Hypothesis\ Space)$$

while improving prediction, unification, transferability, falsifiability, and description efficiency.

Thus:

$$TheoryValue = \Delta Cbit_{eff}^{scientific}$$

12. Methodology: Measuring Cbit

Cbit is a theoretical unit, but it must be approximately measurable.

12.1 Explicit Candidate Sets

In some tasks, candidate structures can be explicitly listed, such as diagnostic hypotheses, scientific hypotheses, or common misconception families.

Then one can estimate:

$$p_i = P(S_i)$$

and compute:

$$H_c = - \sum_i p_i \log_2 p_i$$

12.2 Behavioral Distributions

When internal candidates are not directly observable, behavioral distributions can approximate cognitive entropy: answer distributions, choice distributions, or explanation-path distributions.

12.3 Trajectory Sampling

For complex cognitive processes, intermediate reasoning trajectories can be sampled and treated as approximate observations of Γ_i .

12.4 Error-Family Analysis

Errors are not mere noise. They reveal candidate structures that were competitive. Error families can therefore help reconstruct the possibility space.

12.5 Transfer Tests

If a compression is truly effective, it should transfer to isomorphic or near-isomorphic tasks. Transfer tests can therefore estimate the quality term Q_c .

12.6 Limits of Cbit Measurement

Cbit may not be directly measurable in every system. The claim is not that every cognitive entropy can be precisely measured, but that cognition should be approximable through candidate sets, behavioral distributions, trajectory samples, error families, and transfer tests.

13. Conclusion

This paper proposes a mathematical framework for general cognitive systems.

The cognitive ontology is:

$$\mathcal{P}_c = (\Omega_c, \mathcal{F}_c)$$

where Ω_c is the possibility space and $\mathcal{F}_c$ the constraint field.

Candidate cognitive structures are defined as:

$$S_i = (O_i, R_i, C_i, \Gamma_i)$$

Cognitive entropy is:

$$H_c = - \sum_i p_i \log_2 p_i$$

Cbit is:

$$\Delta Cbit = H_c^{before} - H_c^{after}$$

Effective Cbit is:

$$Cbit_{eff} = \Delta H_c \cdot Q_c$$

The central claim is that cognition is not answer generation, symbolic manipulation, or message transmission. It is effective Cbit compression of a possibility space under a constraint field.

Learning, understanding, reasoning, memory, creativity, misunderstanding, hallucination, scientific research, and education can all be interpreted as different forms of possibility-space compression.

Information theory is not rejected. It is repositioned as a projection of cognitive theory onto message space.

LLMs are not the ontology of this paper and are not used as evidence here. They are only one artificial realization domain where the predictions of the theory may later be tested.

The final formulation is:

A cognitive system is a dynamical system over a possibility space under a constraint field.

Cognition is the effective Cbit compression of that possibility space.

Appendix A: Glossary

Possibility Space: Ω_c , the global domain of candidate possibilities faced by the cognitive system.

Constraint Field: $\mathcal{F}_c$, the multidimensional field of viability and quality constraints acting on the possibility space.

Candidate Cognitive Structure: $S_i = (O_i, R_i, C_i, \Gamma_i)$, a structured candidate element in the possibility space.

Cognitive Entropy: $H_c = - \sum_i p_i \log_2 p_i$, uncertainty over candidate cognitive structures.

Cbit: the compression unit of the candidate cognitive structure possibility space.

Effective Cbit: $Cbit_{eff} = \Delta H_c \cdot Q_c$, cognitive compression weighted by a quality term.

Cognitive Trajectory: $\gamma : [0, T] \rightarrow \Omega_c$, the path of cognitive evolution in possibility space.

Cognitive Action: $\mathcal{A}_c(\gamma)$, total cost of a cognitive trajectory.

Message Projection: $\pi_m : \Omega_c \rightarrow \Omega_m$, a lossy projection from cognitive structures to message space.

Hallucination: the dominance of a wrong but locally closed cognitive trajectory.

Appendix B: LLM Prediction Interface Summary

This paper reports no LLM empirical results. LLMs are retained only as a future empirical interface.

Testable predictions include:

1. LLMs should contain internal proxy quantities for candidate possibility-space compression.
2. High-quality outputs should correspond to higher effective compression rates, not merely higher token confidence.
3. Prompts should alter the unfolding of initial possibility space, not only final answer distributions.
4. Hallucinations should exhibit wrong closure trajectory structures rather than random noise.
5. Effective intervention should alter trajectory advantage structures rather than merely increase answer-token probability.
6. Long-term memory should be evaluated by future Cbit gain rather than textual storage volume.

These predictions are intended to be tested in a separate empirical paper.